

A h-adaptive Entropy-Stable Oscillation-Eliminating Discontinuous Galerkin Method for the Euler Equations on Curvilinear Meshes

Jielin Yang Guosheng Fu

*Department of Applied and Computational Mathematics and Statistics
University of Notre Dame*

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Motivation

- We are interested in robust high-order **Discontinuous Galerkin**(DG) methods for the compressible Euler equations.
- The target problems contain curved geometry, shocks, low-density regions, and multiscale flow structures.
- A practical method should ideally combine:
 - high-order accuracy for smooth solutions,
 - entropy stability,
 - positivity preservation,
 - oscillation control near shocks,
 - and adaptive mesh refinement (AMR).

This talk

- First: propose a entropy-stable DG framework on curvilinear conforming meshes.
- Then: the AMR extension on nonconforming meshes.

We consider the compressible Euler equations

$$\partial_t \mathbf{U} + \nabla \cdot \mathbf{F}(\mathbf{U}) = 0, \quad \mathbf{U} = \begin{pmatrix} \rho \\ \rho \mathbf{u} \\ E \end{pmatrix}.$$

For admissible weak solutions, the physical entropy should satisfy

$$\partial_t \eta(\mathbf{U}) + \nabla \cdot \mathbf{q}(\mathbf{U}) \leq 0.$$

- Entropy stability selects physically relevant solutions after shock formation.
- Positivity of density and pressure is essential for robustness.

Curvilinear DGSEM

Domain Ω is decomposed into curvilinear quadrilateral elements, each mapped from the reference element $\Omega_{\text{ref}} = [-1, 1]^2$ via $\mathbf{x}^e(\xi, \eta)$.

Metric terms: $x_\xi^e, y_\eta^e, x_\eta^e, y_\xi^e$; **Jacobian:** $J^e = x_\xi^e y_\eta^e - x_\eta^e y_\xi^e$;

Contravariant fluxes:

$$\tilde{f}_k^e = y_\eta^e f_k - x_\eta^e g_k, \quad \tilde{g}_k^e = -y_\xi^e f_k + x_\xi^e g_k$$

The Euler equations on Ω_{ref} read

$$J^e \partial_t U_k^e + \partial_\xi \tilde{f}_k^e + \partial_\eta \tilde{g}_k^e = 0.$$

DG weak form:

$$\int_{\Omega_{\text{ref}}} J^e \partial_t U_k^e \hat{\Phi} - \int_{\Omega_{\text{ref}}} \tilde{f}_k^e \partial_\xi \hat{\Phi} - \int_{\Omega_{\text{ref}}} \tilde{g}_k^e \partial_\eta \hat{\Phi} + \int_{\partial\Omega_{\text{ref}}} (\tilde{f}_k^{e,*}, \tilde{g}_k^{e,*}) \cdot \hat{n} \hat{\Phi} d\hat{s} = 0.$$

DG strong form:

$$\int_{\Omega_{\text{ref}}} J^e \partial_t U_k^e \hat{\Phi} + \int_{\Omega_{\text{ref}}} (\partial_\xi \tilde{f}_k^e + \partial_\eta \tilde{g}_k^e) \hat{\Phi} - \int_{\partial\Omega_{\text{ref}}} [(\tilde{f}_k^e - \tilde{f}_k^{e,*}, \tilde{g}_k^e - \tilde{g}_k^{e,*}) \cdot \hat{n}] \hat{\Phi} d\hat{s} = 0.$$

Approximation space:

$$\mathcal{Q}_N(\Omega_{\text{ref}}) = \text{span}\{\varphi_i(\xi)\varphi_j(\eta)\}_{i,j=0}^N,$$

where $\{\varphi_i\}$ are the Lagrange polynomials at the $N + 1$ Gauss-Lobatto nodes.

Semi-discrete DGSEM:

$$\begin{aligned} w_{p_1} w_{p_2} J_{p_1 p_2}^e (\partial_t U_k^e)_{p_1 p_2} + w_{p_1} w_{p_2} \sum_i D_{p_1 i} (\tilde{f}_k^e)_{i p_2} + w_{p_1} w_{p_2} \sum_j D_{p_2 j} (\tilde{g}_k^e)_{p_1 j} \\ + \text{interface flux integrals} = 0. \end{aligned}$$

Local Lax–Friedrichs interface flux:

$$\begin{aligned} F_k^* \cdot \mathbf{n} &= \frac{1}{2} [\mathbf{n} \cdot F_k(U^L) + \mathbf{n} \cdot F_k(U^R)] - \frac{\alpha}{2} [(U_k)^R - (U_k)^L]. \\ \alpha &= \max\{|\mathbf{u}^L \cdot \mathbf{n}| + c^L, |\mathbf{u}^R \cdot \mathbf{n}| + c^R\}. \end{aligned}$$

Key idea [Tadmor 1987]: replace the pointwise nonlinear volume flux by entropy-conservative two-point fluxes.

In the standard DGSEM,

$$\sum_i D_{p_1,i}(\tilde{f}_k^e)_{i,p_2} \implies 2 \sum_i D_{p_1,i}(\tilde{f}_k^{e,\#})_{(i,p_1),p_2}.$$

Tadmor's entropy-conservative flux

A symmetric, consistent two-point flux $F^\#(U^L, U^R)$ is entropy conservative if

$$(\mathbf{V}^R - \mathbf{V}^L) \cdot F^\#(U^L, U^R) = \psi^R - \psi^L,$$

where $\mathbf{V}$ is the entropy variable and ψ is the entropy potential flux.

Reference: E. Tadmor, *The numerical viscosity of entropy stable schemes for systems of conservation laws. I*, Math. Comp., 49 (1987), 91–103.

Chandrashekar flux [2013] is entropy conservative and kinetic-energy preserving:

$$f_1^{\#} = \rho^{\log} \{\{u\}\}, \quad f_2^{\#} = \rho^{\log} \{\{u\}\}^2 + \hat{p}, \quad f_3^{\#} = \rho^{\log} \{\{u\}\} \{\{v\}\}, \quad f_4^{\#} = \rho^{\log} \{\{u\}\} \hat{h}.$$

Metric averaging: on curved meshes, the contravariant two-point flux must also average the metric terms:

$$(\tilde{f}_k^{a,\#})_{(j,p_1),p_2} = \{\{y_\eta^e\}\}_{(j,p_1),p_2} (f_k^{\#})_{(j,p_1),p_2} - \{\{x_\eta^e\}\}_{(j,p_1),p_2} (g_k^{\#})_{(j,p_1),p_2}.$$

Key point

Symmetric metric averaging is the mechanism that makes the entropy proof work on curvilinear meshes.

Reference: P. Chandrashekar, *Kinetic energy preserving and entropy stable finite volume schemes for compressible Euler and Navier-Stokes equations*, Commun. Comput. Phys., 14 (2013), 1252–1286.

Entropy-Stable DGSEM

To prove **global discrete entropy stability**, two algebraic ingredients are essential:

Summation-by-Parts (SBP)

Derivative matrix:

$$D_{ij} = \varphi_j'(\xi_i), \quad i, j = 0, \dots, N,$$

where $\{\xi_i\}_{i=0}^N$ are the Gauss-Lobatto nodes, and $\{w_i\}_{i=0}^N$ are the quadrature weights, then we have:

$$w_i D_{ij} + w_j D_{ji} = \delta_{N,i} \delta_{N,j} - \delta_{0,i} \delta_{0,j}.$$

This is the discrete analogue of integration by parts.

Discrete Metric Identities [Kopriva 2006]

When $x^e \in \mathcal{Q}_N(\Omega_{\text{ref}})$, the LGL differentiation satisfies

$$\sum_{p_2} D_{q_2, p_2}(y_\xi)_{q_1, p_2} = \sum_{p_1} D_{q_1, p_1}(y_\eta)_{p_1, q_2}, \quad \forall q_1, q_2.$$

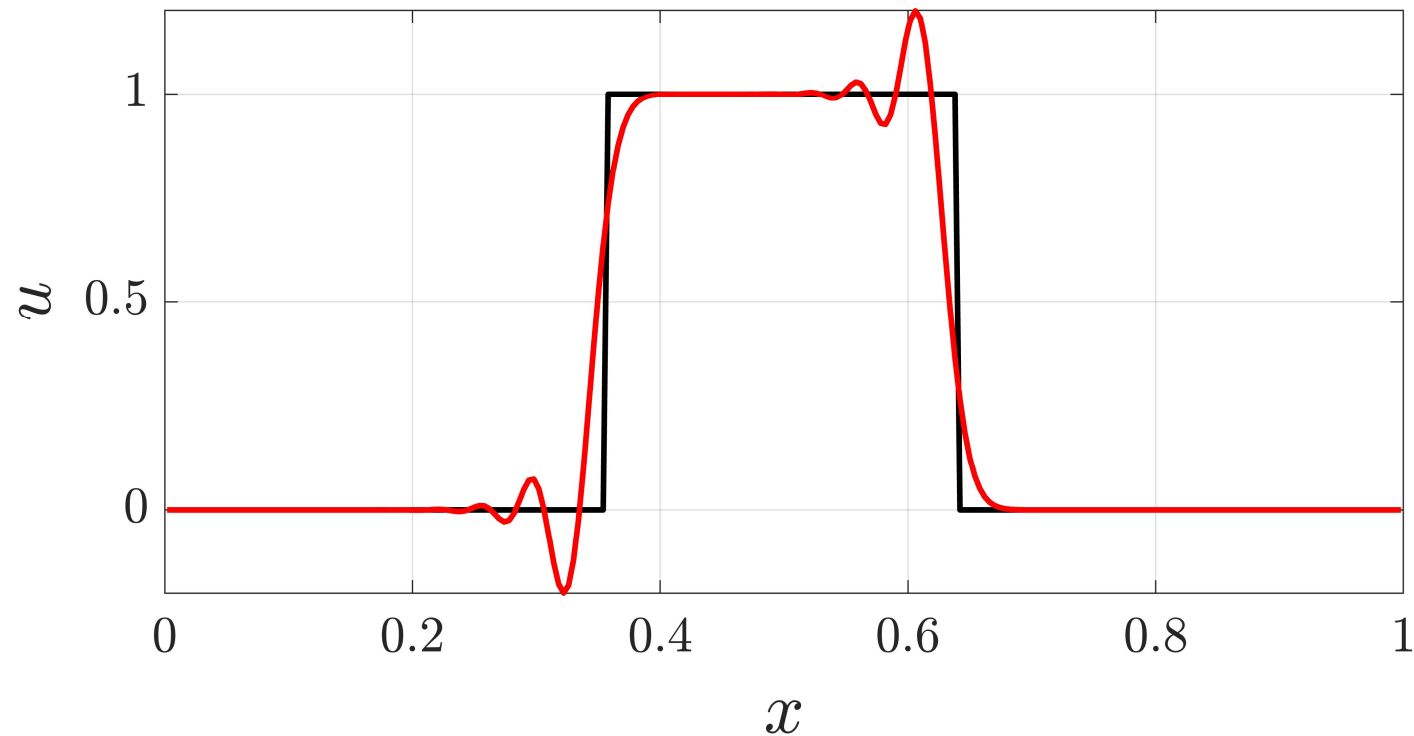
Theorem (Global discrete entropy stability)

Under periodic boundary conditions, if the interface flux is entropy-stable:

$$(\mathbf{V}^R - \mathbf{V}^L) \cdot [F^*(U^L, U^R) \cdot \mathbf{n}] - (\psi^R - \psi^L) \cdot \mathbf{n} \leq 0,$$

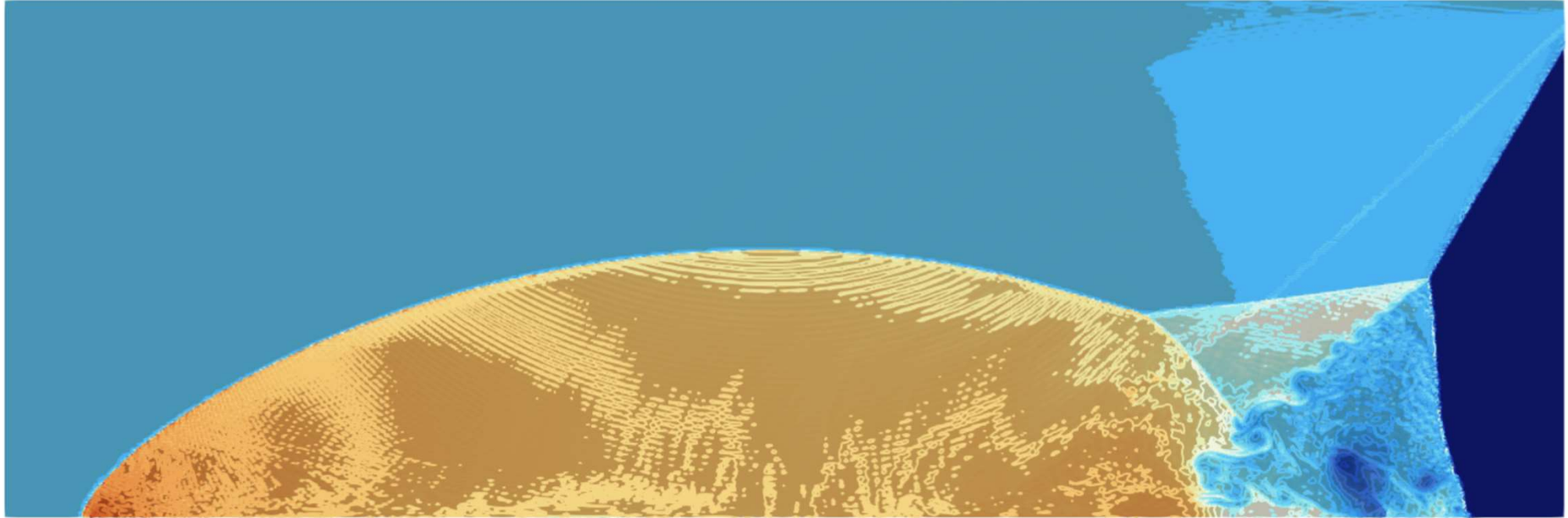
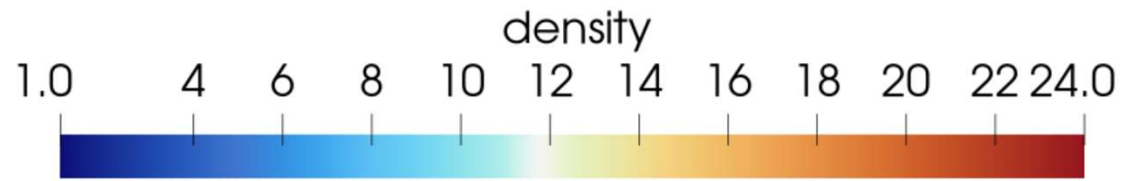
then the curvilinear ES-DGSEM satisfies:

$$\sum_e \sum_{p_1, p_2} w_{p_1} w_{p_2} J_{p_1 p_2}^e \left(\partial_t \eta(\mathbf{U}_N^e) \right)_{p_1 p_2} \leq 0.$$



Numerical oscillation near the discontinuous points

Oscillation-Eliminating (OE)



Double Mach reflection problem.
Dirty contours are caused by numerical oscillation.

Oscillation-Eliminating (OE)

Entropy stability alone does not eliminate post-shock oscillations.

- High-order DG methods typically require additional nonlinear stabilization near discontinuities.
- Common choices include TVB limiters, WENO reconstructions, and artificial viscosity.
- Our choice is a **filter-type** stabilization: the oscillation-eliminating (OE) procedure, following the recent OEDG work of K. Wu (Math. Comp., 94 (2025), 1147–1198.).
- In our work, we further improved this idea and extended it to entropy-stable DGSEM on curvilinear meshes.

Main idea

Dampening the higher-order terms properly, while preserving the cell average.

Reference: M. Peng, Z. Sun, and K. Wu, *OEDG: Oscillation-eliminating discontinuous Galerkin method for hyperbolic conservation laws*, Math. Comp., 94 (2025), 1147–1198.

Oscillation-Eliminating (OE)

The OE procedure is defined through a **pseudo-time ODE**: for each component,

$$\frac{d}{d\hat{t}} \int_{\Omega^e} U_{\sigma}^{e,k} v \, d\mathbf{x} + \sum_{m=0}^N \delta_m^{(e)} \int_{\Omega^e} (U_{\sigma}^{e,k} - P^{m-1} U_{\sigma}^{e,k}) v \, d\mathbf{x} = 0.$$

where P^m is the projection operator, and $\delta_m^{(e)}$ is the damping coefficient.

On Cartesian meshes, one can use an **orthogonal modal basis**

$$U_{\sigma}^{e,k} = \sum_{ij} (\hat{U}_{\sigma}^{e,k})_{ij} \Psi_{ij},$$

which gives the exact solution

$$\mathcal{F}_{\Delta t}(U_h^{e,k}) = \boxed{(\hat{U}_h^{e,k})_{0,0} \Psi_{0,0}} + \sum_{\max(i,j) \geq 1} e^{-\Delta t \sum_{m=0}^{\max(i,j)} \delta_m^{(e)}} (\hat{U}_h^{e,k})_{ij} \Psi_{ij}.$$

Oscillation-Eliminating (OE)

On general curved meshes, an element-local orthogonal modal basis is not readily available.

We use the nested projections

$$u = P^0 u + \sum_{k=1}^N (P^k u - P^{k-1} u),$$

where the L^2 projection P^k onto $\mathcal{Q}_k(\Omega^e)$ is defined by

$$\int_{\Omega^e} (P^k w - w) v \, d\mathbf{x} = 0, \quad \forall v \in \mathcal{Q}_k(\Omega^e).$$

This leads to an exact closed-form OE filter on curvilinear elements:

$$U_{\sigma}^{e,r} = P^0(U_h^{e,r}) + \sum_{k=1}^N e^{-s\Delta t \sum_{n=1}^k \mathcal{J}_{\sigma}^{(n)}} (P^k(U_h^{e,r}) - P^{k-1}(U_h^{e,r})).$$

The damping coefficients are defined by

$$\delta_m^{(e)} = \frac{\beta_e}{h_e} \sum_{f \in \partial \Omega^e} \sigma_m^{(f)}(\mathbf{U}_h^e), \quad \beta_e = \|\bar{\mathbf{u}}\| + \bar{\epsilon}.$$

For each face f ,

$$\sigma_m^{(f)}(\mathbf{U}_h^e) = \max_{1 \leq k \leq 4} \sigma_m^{(f)}(U_h^{e,k}),$$

with

$$\sigma_m^{(f)}(U_h^{e,k}) = \frac{(2m+1)h_e^m}{2(2N-1)m!} \left[\sum_{n \leq m} \frac{\frac{1}{|f|} \int_f |[\partial_x^n U_h^{e,k}]_f| dS}{\|U_h^{e,k} - \bar{U}_h^{e,k}\|_{L^\infty(\Omega^e)}} + \sum_{n \leq m} \frac{\frac{1}{|f|} \int_f |[\partial_y^n U_h^{e,k}]_f| dS}{\|U_h^{e,k} - \bar{U}_h^{e,k}\|_{L^\infty(\Omega^e)}} \right].$$

The zeroth-order coefficient is especially important:

$$\sigma_0^{(f)}(U_h^{e,r}) = \frac{1}{2(2N-1)} \frac{\frac{1}{|f|} \int_f |[[U_h^{e,r}]]_f| dS}{\|U_h^{e,r} - \bar{U}_h^{e,r}\|_{L^\infty(\Omega^e)}}.$$

It leads to the element-wise **shock indicator**:

$$\mathcal{I}^{(e)}(\mathbf{U}_h^e) = \sum_{f \in \partial\Omega^e} \sigma_0^{(f)}(\mathbf{U}_h^e).$$

- Elements with $\mathcal{I}^{(e)} > C$ are marked as troubled.
- Only troubled elements are filtered; smooth elements are left unchanged.
- The same indicator is later reused for adaptive mesh refinement.

Algorithmic workflow

- ➊ Advance the entropy-stable DGSEM with strong-stability-preserving third order Runge-Kutta(**SSP-RK3**).
- ➋ At each stage, compute the shock indicator $\mathcal{I}^{(e)}$.
- ➌ Mark troubled elements and apply the **OE filter** only there.
- ➍ Apply the **Zhang–Shu positivity-preserving limiter**.

What this gives

- provable entropy stability,
- positivity preservation,
- high-order accuracy for smooth solutions,
- effective oscillation control near shocks.

Double Mach Reflection

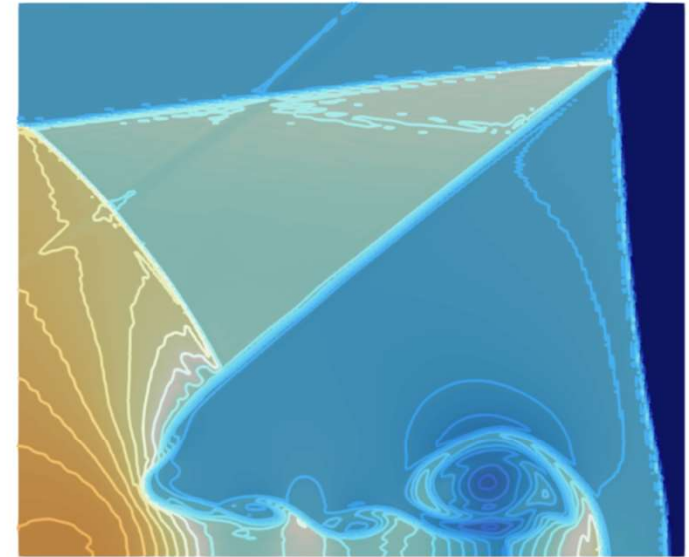
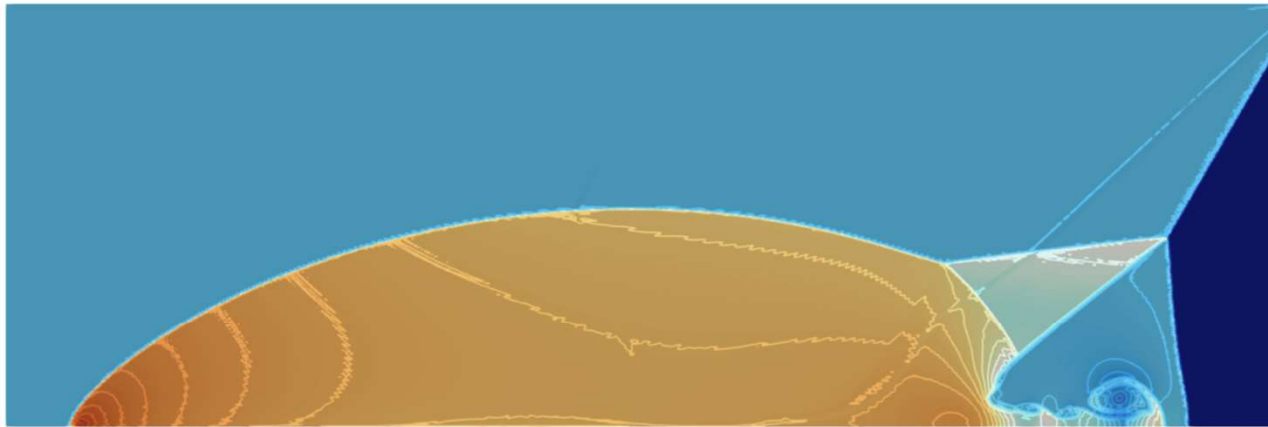


Total number of meshes: 240×960 ; Cartesian structured elements;

Double Mach Reflection

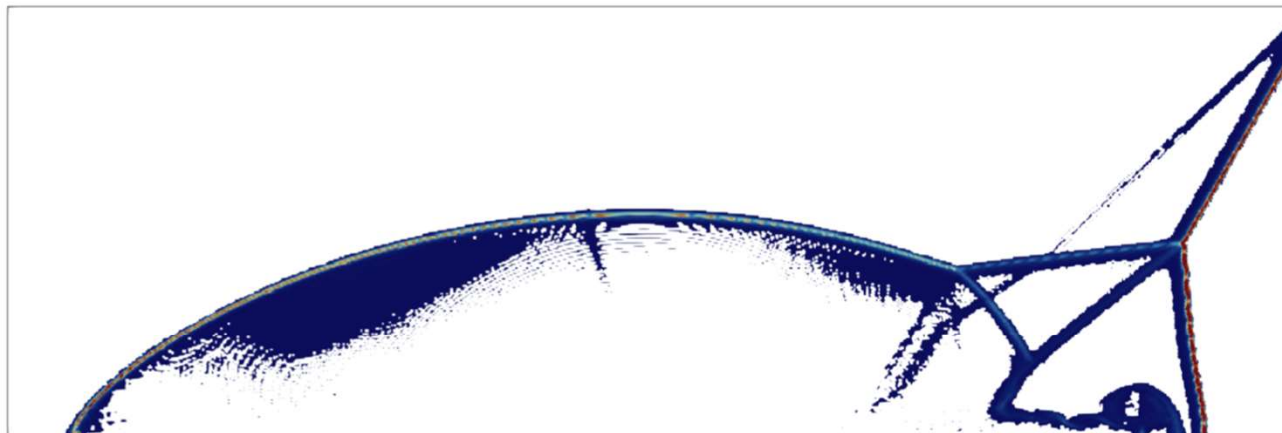
density

1.0 4 6 8 10 12 14 16 18 20 22 24.0



shock indicator

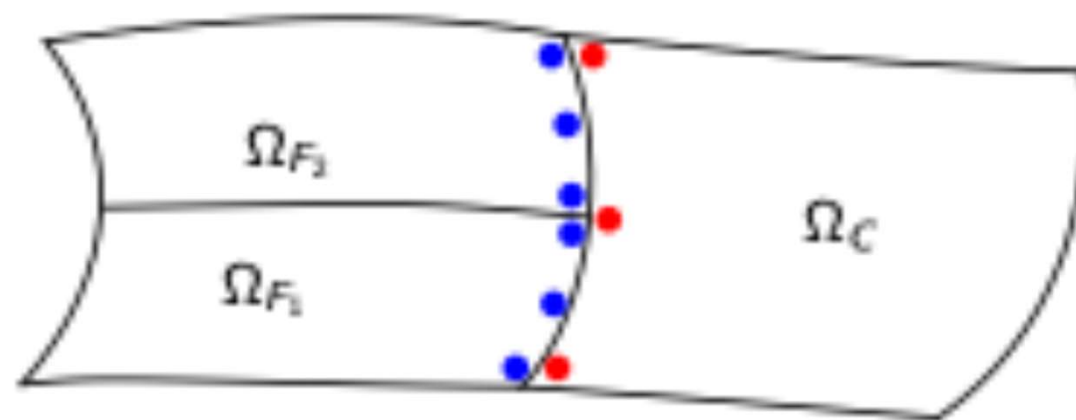
0.00 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5.00



2D Mach 3 Flow Past Cylinder

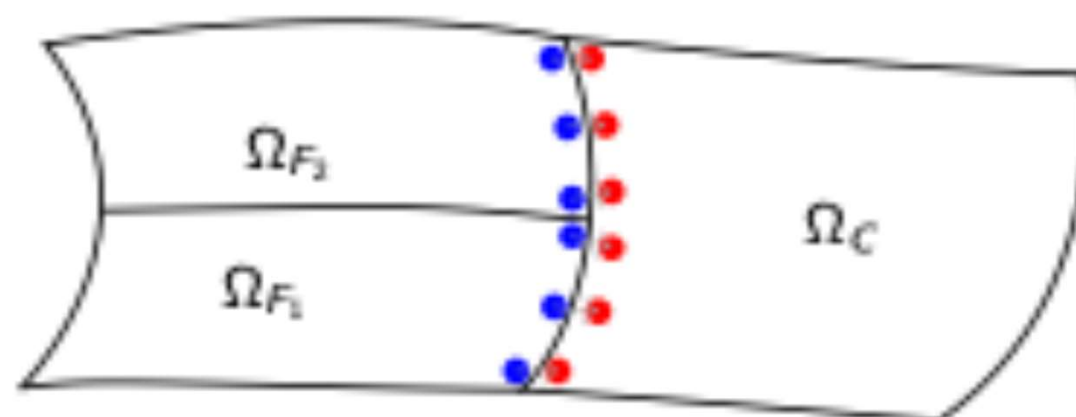


Total number of meshes: 126,082; Curvilinear quadrilateral elements;



- On a nonconforming interface, the fine and coarse trace nodes do not match one-to-one.
- Standard conforming interface flux pairing is no longer available.
- As a result, the usual positivity-preserving and entropy-stability arguments break down.

Mortar-Based Flux: Geometric Picture



- Blue points: fine/mortar evaluation points on the two fine elements.
- Red points: coarse-side states **interpolated** to the same mortar locations.
- The numerical flux is computed pointwise on the mortar nodes, then projected back to the coarse trace.

Mortar-Based Flux

The mortar-based flux proceeds as follows:

- 1 Interpolate the coarse-side state values to the fine/mortar points:

$$\tilde{\mathbf{U}}_i^{C,k} = \sum_j (P_{C \rightarrow F_k})_{ij} \mathbf{U}_j^C.$$

- 2 Construct a one-to-one interface flux on the fine/mortar points, and then project it back to the coarse side:

$$\text{On fine side: } \mathbf{F}_i^{F_k,*} \cdot \mathbf{n}_i^{F_k} := \mathbf{F}^*(\mathbf{U}_i^{F_k}, \tilde{\mathbf{U}}_i^{C,k}) \cdot \mathbf{n}_i^{F_k},$$

$$\text{On coarse side: } \mathbf{F}_j^{C,*} \cdot \mathbf{n}_j^C := -2 \sum_{k=1}^2 \sum_{i=0}^N (P_{F_k \rightarrow C})_{ji} \mathbf{F}^*(\mathbf{U}_i^{F_k}, \tilde{\mathbf{U}}_i^{C,k}) \cdot \mathbf{n}_i^{F_k}.$$

Properties

High-order accurate and positivity preserving, but no entropy-stability proof.

Instead of forming a mortar state first, we can assemble the interface flux pairwise:

$$F_j^L = \sum_i (P_{C \rightarrow F})_{ji} F^*(\mathbf{U}_j^L, \mathbf{U}_i^R), \quad F_i^R = \sum_j (P_{F \rightarrow C})_{ij} F^*(\mathbf{U}_j^L, \mathbf{U}_i^R).$$

- This construction keeps the same pairwise coarse–fine coupling.
- It empirically retains high-order accuracy.
- However, the interpolation matrix may contain negative entries.

Consequence

Because of the sign changes in the projection/interpolation matrices, we currently have no proof of positivity preservation or entropy stability for the interpolation-based flux.

Entropy-Stable Flux

The entropy-stable flux modifies the pairwise construction when the interpolation weight changes sign.

Schematically,

$$F_j^L = \sum_i (P_{C \rightarrow F})_{ji} \begin{cases} F^*(\mathbf{U}_j^L, \mathbf{U}_i^R), & (P_{C \rightarrow F})_{ji} \geq 0, \\ F^*(\mathbf{U}_i^R, \mathbf{U}_j^L), & (P_{C \rightarrow F})_{ji} < 0, \end{cases}$$

and similarly for the coarse-side contribution.

- The sign-aware swapping restores the algebra needed in the entropy proof.
- It also fits into the positivity-preserving framework.
- The price is a loss of full high-order consistency.

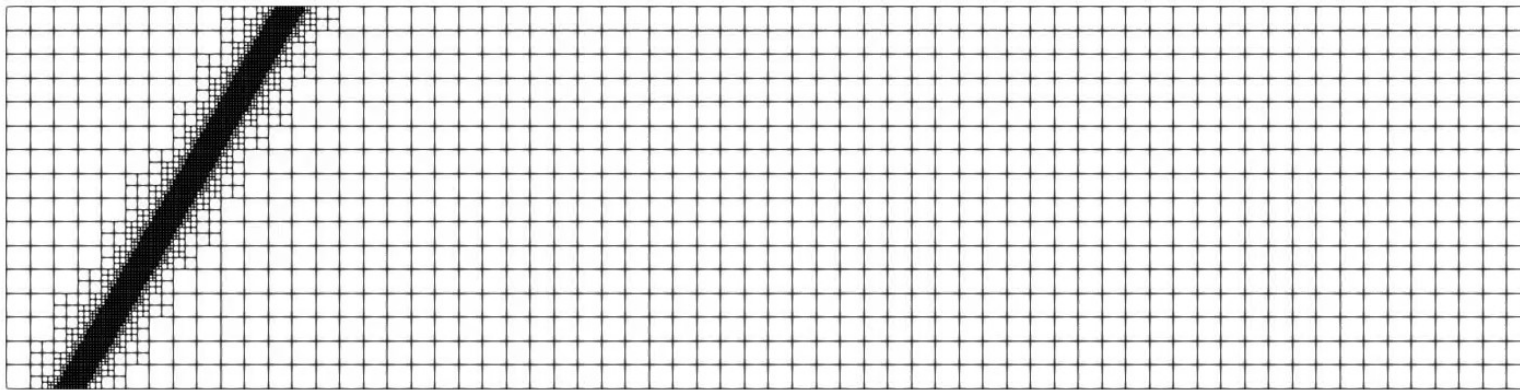
Properties

Entropy stable and positivity preserving, but only first-order accurate in general.

Flux	High-order	PP proof	ES proof
Mortar-based flux	yes	yes	no
Interpolation-based flux	yes	no	no
Entropy-stable flux	no	yes	yes

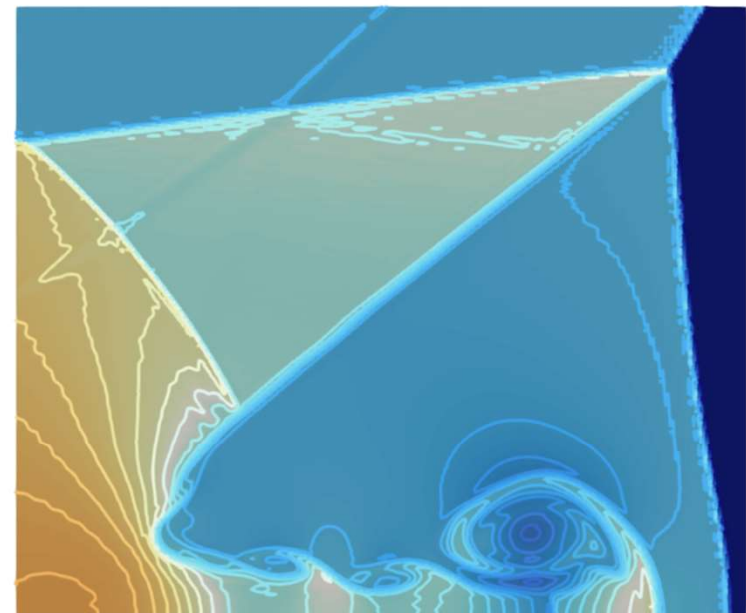
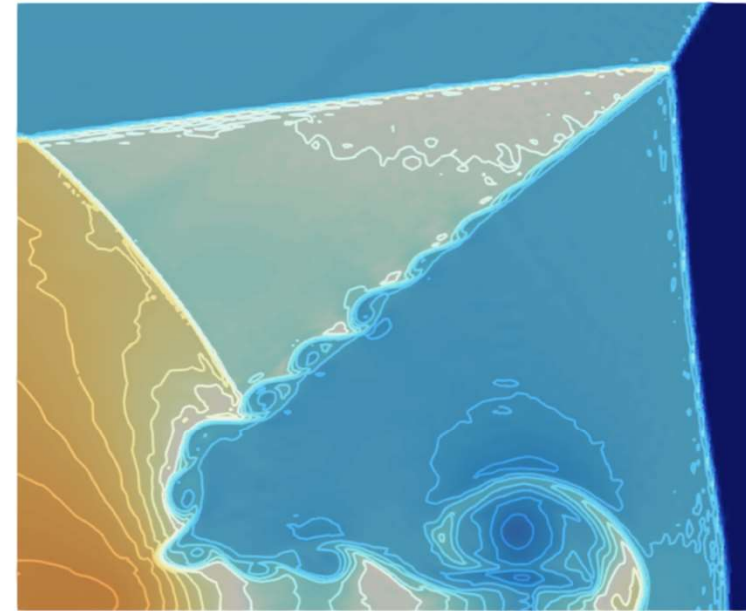
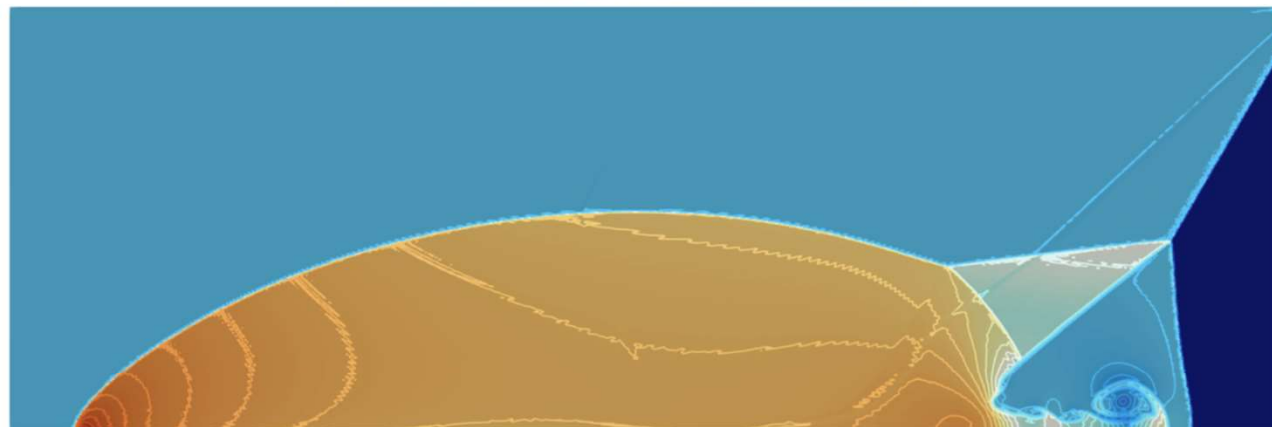
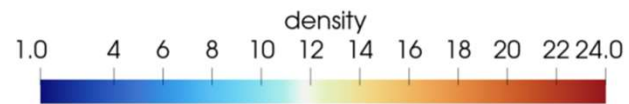
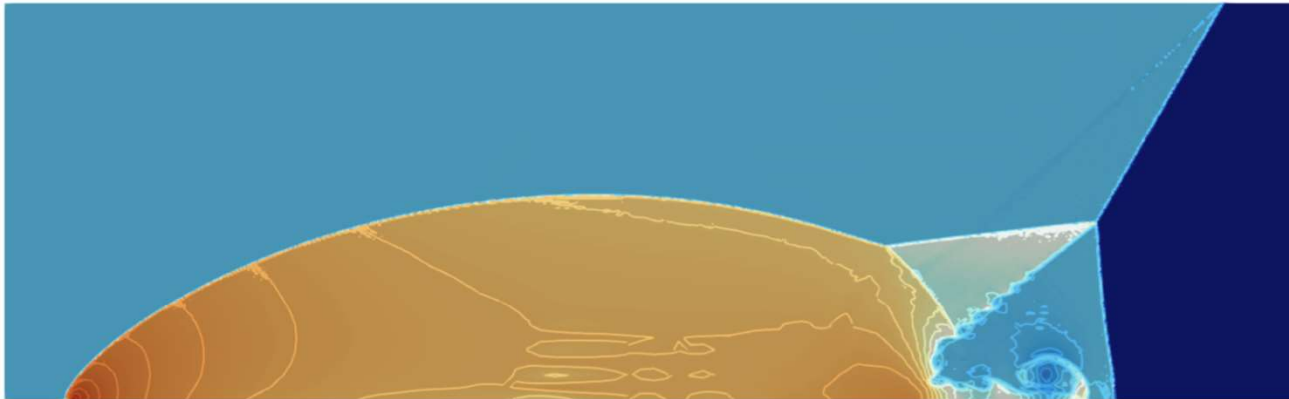
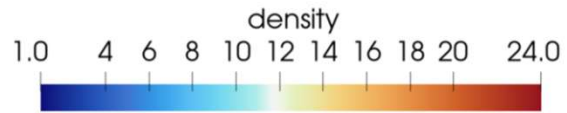
- We use the zeroth- $\mathcal{O}E$ damping coefficient as the AMR indicator.
- We also developed AMR data transfer, positivity-preserving transfer, and mortar-aware limiting.
- The full AMR solver is implemented on both Cartesian and curvilinear meshes.

Double Mach Reflection with AMR



Initial number of meshes: 16×64 ; Max refinement level 5;
Max number of elements 19,720;

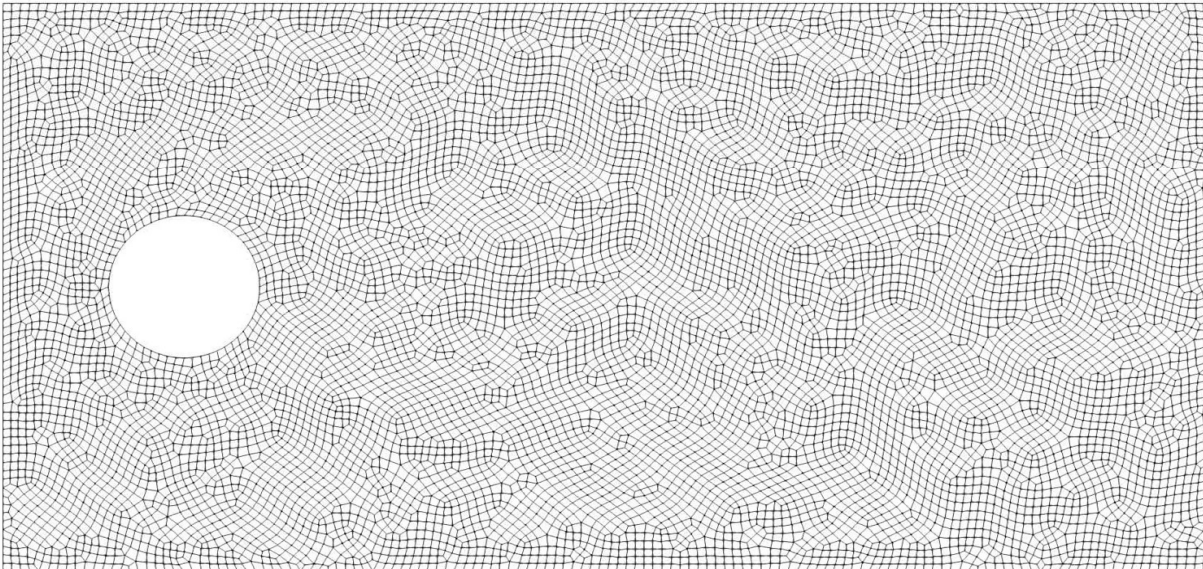
Double Mach Reflection with AMR



2D Mach 3 Flow Past Cylinder with AMR



Initial number of meshes: 10,397;
Max refinement level: 3;
Max number of elements: 47,681;



Take-home messages

- A conforming curvilinear entropy stable DG framework with provable entropy stability.
- Oscillation-eliminating(OE) procedure to control the oscillation near shocks.
- AMR on nonconforming meshes with three interface fluxes.
- All code implementations are based on the open-source library MFEM(developed by LLNL).

Open issue

Provable entropy stability, provable positivity preservation, and full high-order accuracy are still not simultaneously achieved in one nonconforming flux.

Thank you!

jyang38@nd.edu gfu@nd.edu

